

YIFAN WANG

Boston, MA | wang.yifan25@northeastern.edu | [Personal Page](#) | [Linkedin](#) | [Google scholar](#)

EDUCATION

Northeastern University, Boston, MA

Sep. 2023 – Jun. 2028 (Expected)

Ph.D in Computer Engineering under supervision of [Prof. Yun \(Raymond\) Fu](#)

Research Interests: Video Generation Model, Model Efficiency, Computer Vision

Shandong University, Jinan, Shandong, P.R.China

Sep. 2018 – Jun. 2022

Bachelor in Mathematics and applied mathematics

RESEARCH INTERESTS

My research focuses on the post-training and alignment of video diffusion models, with a particular emphasis on reward finetuning methods and RLHF-based policy optimization. I aim to enhance the spatiotemporal consistency and instruction-following capabilities of large-scale video generative systems.

PUBLICATIONS

Conferences:

- **Yifan Wang**, Yanyu Li, Gordon Guocheng Qian, Sergey Tulyakov, Yun Fu, Anil Kag. “Diffusion-DRF: Free, Rich, and Differentiable Reward for Video Diffusion Fine-Tuning”. *Under Review*. [\[Project page\]](#)
- **Yifan Wang**, Xu Ma, Yitian Zhang, Yizhou Wang, Zhongruo Wang, Sung Cheol Kim, Vahid Mirjalili, Visya Renganathan, Yun Fu. “GmNet: Revisiting Gating Mechanisms from A Frequency View”. **Accepted by *ICLR*, 2026**. [\[Paper\]](#)
- **Yifan Wang**, Yun Fu. “Understanding, Abstracting and Checking: Evoking Complicated Multimodal Reasoning in LMMs”. **Accepted by *ACL Findings*, 2026**
- **Yifan Wang**, Yi Xu, Yue Bai, Yun Fu. “Hyperbolic Few-shot Open-set Recognition”. Under review

Journals:

- **Yifan Wang**, Lin Zhang, Ran Song, Lin Ma, Wei Zhang. “Exploiting Inter-Sample Affinity for Knowability-Aware Universal Domain Adaptation”. **Accepted by *International Journal of Computer Vision (IJCV)*, 2023**. [\[Paper\]](#)
- Lin Zhang, Ran Song, **Yifan Wang**, Qian Zhang, Xiaolin Wei, Lin Ma, Wei Zhang. “Dual-Graph Contrastive Learning for Unsupervised Person Re-Identification”. **Accepted by *IEEE Transactions on Multimedia (TMM)*, 2023** [\[Paper\]](#)
- Lin Zhang, **Yifan Wang**, Ran Song, Mingxin Zhang, Xiaolei Li, Wei Zhang. “Neighborhood-Aware Mutual Information Maximization for Source-Free Domain Adaptation” **Accepted by *IEEE Transactions on Multimedia (TMM)*, 2023** [\[Paper\]](#)

RESEARCH EXPERIENCE

[Creative Vision@Snap Inc.](#), *Research Intern*, Santa Monica, CA

Jun. 2025 - Now

Mentor: Anil Kag, Gordon Guocheng Qian, Yanyu Li and Sergey Tulyakov

- **Overview:** During the internship in Snap, I’m working on post-training of video diffusion. Targeting effective and efficient RL method to improve the general capability of video diffusion models.
- **TC-GRPO: VLM Gradients Provide Token-Level Credit for GRPO in Video Generation Model**
Research Brief: Developed a token-credit reinforcement learning framework for text-to-video generation that improves GRPO by using VLM-derived gradient signals to identify reward-sensitive video frames. The method decouples scalar reward estimation from credit assignment: reward models provide the advantage, while pretrained VLM gradients produce frame-level weights that reweight the log-probability ratio during policy optimization.
- **Diffusion-DRF: Differentiable Reward Flow for Video Diffusion Fine-Tuning**
Research Brief: We propose Diffusion-DRF, a free, rich, and differentiable reward framework for video diffusion fine-tuning. Diffusion-DRF employs a frozen, off-the-shelf Vision-Language Model (VLM) as the critic, eliminating

the need for reward model training. Instead of relying on a single scalar reward, it decomposes each user prompt into multi-dimensional questions with freeform dense VQA explanation queries, yielding information-rich feedback.

[Smile Lab](#)@Northeastern University, *Ph.D student*, Boston, MA

Sep. 2023 - Now

Advisor: Prof. Yun Raymond Fu

- **Overview:** In Smile Lab, I'm exploring different topics to extend my view of CV. I've worked on light-weight models, VLM Agent and few-shot recognition.
- **GmNet: Revisiting Gating Mechanisms from A Frequency View.**
Research Brief: We systematically explore the effect of gating mechanisms on the training dynamics of neural networks from a frequency perspective. Leveraging insights on GLUs, we propose a Gating Mechanism Network (GmNet), a lightweight model designed to efficiently utilize the information of various frequency components. GmNet achieves impressive performance in terms of both effectiveness and efficiency in the image classification task.
- **Evoking Complicated Multimodal Reasoning in LMMs.**
Research brief: We present a prompting method named UnAC (Understanding, Abstracting, and Checking), to synergize reasoning for complicated problems in the multimodal context of LMMs, such as GPT-4V, Gemini-provision and LLaVA. UnAC can improve the understanding of the image capturing more details and making better reasoning of complicated multimodal questions.
- **Hyperbolic Few-shot Open-set Recognition.**
Research brief: Propose a hyperbolic framework based on a Hyperbolic Conditional Variational Autoencoder to address the issues that existing approaches suffer from bias due to data scarcity and leading to poorly represented hierarchical class features.

[Vsislab](#)@Shandong University, *Research Assistant*, Jinan, Shandong, P.R.China

Mar, 2021 – Mar, 2023

Advisor: Prof. [Wei Zhang](#)

- **Exploiting Inter-Sample Affinity for Knowability-Aware Universal Domain Adaptation.**
Research Brief: Proposed a novel UniDA framework to fully exploring the inter-sample affinity between known and unknown samples. Specifically, we introduced a knowability-based labeling scheme which can be divided into two steps: (1) Knowability-guided detection of known and unknown samples based on the intrinsic structure of the neighborhoods of samples. (2) Label refinement based on neighborhood consistency to relabel the target samples.
- **Neighborhood-Aware Mutual Information Maximization for Source-Free Domain Adaptation**
Research brief: Proposed the neighborhood-aware mutual information maximization (NAMI), which maximizes the mutual information (MI) between the representations of target domain and the corresponding neighborhood. Incorporated the consistency constraint between different views into the method to encourage the model to be robust to changes in the sample space and thus further alleviate the problem of domain gap.
- **Dual-Graph Contrastive Learning for Unsupervised Person Re-Identification**
Research brief: Proposed an effective contrastive learning framework called dual-graph contrastive learning for the fully unsupervised person re-ID, which can fully utilize the intrinsic structure of the unlabeled data and better reduce the high intra-class variance. Proposed a camera-aware augmentation scheme that exploits the semantic information between samples and their corresponding camera-level centroids to relieve the domain gap between cameras.

WORK EXPERIENCE

Snap Inc., Santa Monica, CA, USA

Jun, 2025 – Now

- **Mentor:** Yanyu Li
- **Position:** Research Internship
- **Responsibilities:** Engaging the research projects in the post-training of video generation, the primary research domain is RL of video different models.

Northeastern University, Boston, MA, USA

Jun, 2024 – Aug, 2024

- **Supervisor:** Prof. [Yun Raymond Fu](#)

- **Position:** Research Assistant (Full-time)
- **Responsibilities:** Engaging the research projects in computer vision, the primary research domain is Efficient Vision Transformer.

Shandong University, Jinan, Shandong, P.R.China

Mar, 2021 – Mar, 2023

- **Supervisor:** Prof. [Wei Zhang](#)
- **Position:** Research Assistant (Full-time)
- **Responsibilities:** Engaging the research projects in computer vision, the primary research domain including Domain Adaptation, Self-supervised learning and Unsupervised Learning

SKILLS AND INTERESTS

- *Computer Skills:* Python, Matlab
- *Tools and library:* Pytorch, Numpy, Scikit-learn.